Mobicom 4/5G Network Insight & Planning Suggestion

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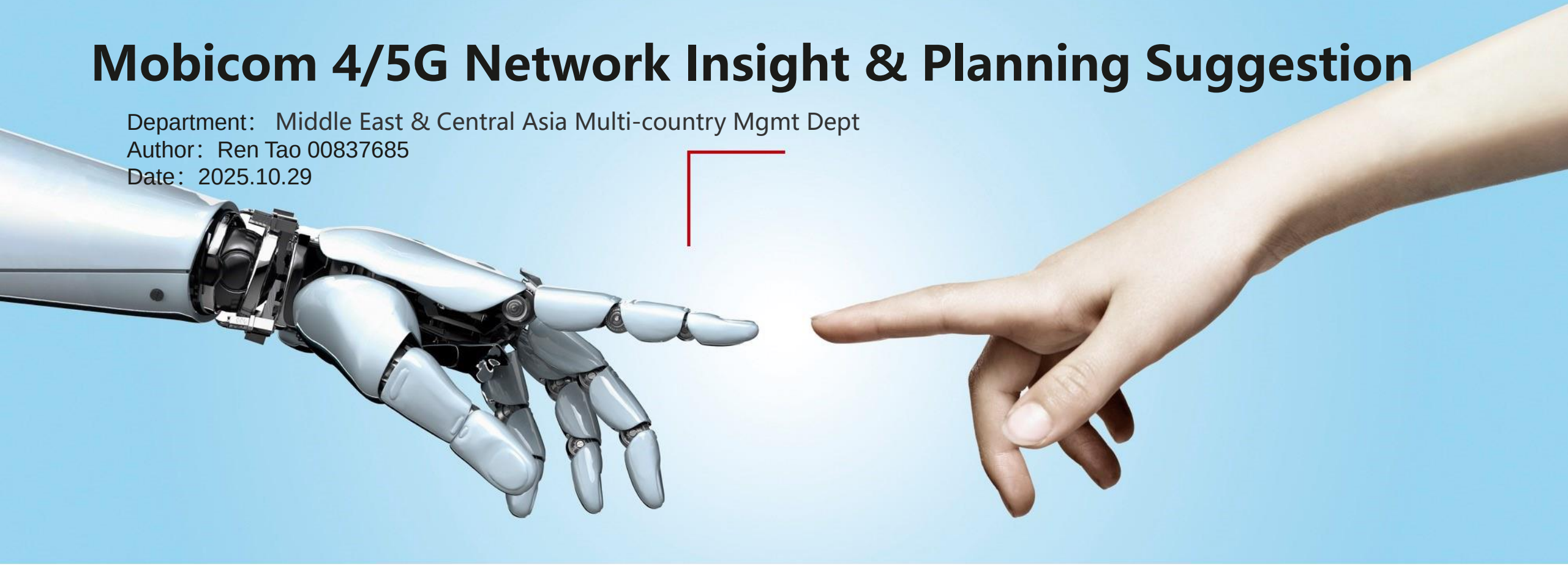
Date：2025.10.29



Security Level:

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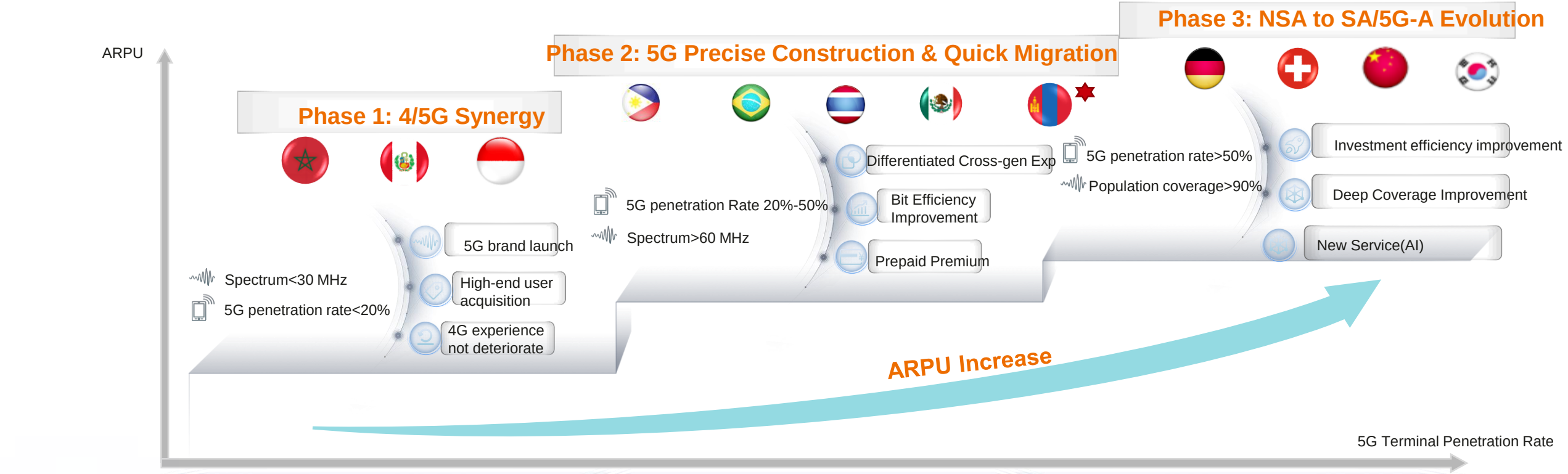
- Industry Trend
- 5G Network Insight
- 4G Network Insight
- Network Planning Suggestion

Network Construction Pace Matches Resource, Fully Releasing 5G Dividends



	Phase 1: 4/5G Synergy	Phase 2: 5G Precise Construction & Quick Migration	Phase 3: NSA to SA/5G-A Evolution
Efficient Network Construction	LNR Dual-Usage	Traffic-based Value Area Selection (5G City)	NSA->SA\5G-A Efficient Evolution
Precise Migration	Three-high Users (Terminal, DOU, ARPU)	New High Potential User	SA User Increase
Prosperous Ecology	Introduction of 5G Entry-level Terminals	5G Terminal: OEM->ODM Content: Long Video->Short video, Short Drama	SA Enabled by Default, RedCap IoT, AI Service

Huawei Proprietary - Restricted Distribution

5G Traffic Offload Improving

Marketing | **Network**

$$\frac{\text{5G Traffic}}{\text{4\&5G Traffic}} = \frac{\text{5G Device Users}}{\text{4\&5G Device Users}} \times \frac{\text{5G Unlock Device Users}}{\text{5G Device Users}} \times \frac{\text{5G User DOU}}{\text{4G User DOU}} \times \frac{\text{5G Traffic Users}}{\text{5G Unlock Device Users}} \times \frac{\text{5G Traffic of 5G Users}}{\text{4\&5G Traffic of 5G Users}}$$

$$\text{5G Traffic Offload} = \text{5G Device Penetration} \times \text{5G Device Unlock Rate} \times \text{5G DOU/4G DOU} \times \text{5G Unlock Access Rate} \times \text{5G Camping Ratio}$$

	5G Traffic Offload	5G Device Penetration	5G Device Unlock Rate	5G DOU/4G DOU	5G Unlock Access Rate	5G Camping Ratio
	12.8%	50%	69%	1.8	48%	43%
2026 Target:	30%	55%	80%	2	60%	60%

Business-Network Synergy

Marketing

- Precision marketing to eliminate user concerns
- Business innovation to attract/stimulate user interest

Network

- Enhance 5G coverage and device-network-tariff compatibility
- Solidify the 5G user experience and deliver cross-gen Experiece

Value Oriented Planning

High Value Area

Mid-Low Valu Area

- Build continuous 5G, 5G User always stay in 5G
- Stimulate 5G User traffic with Cross-gen Experience

8T->MM; DAS->DIS; ...

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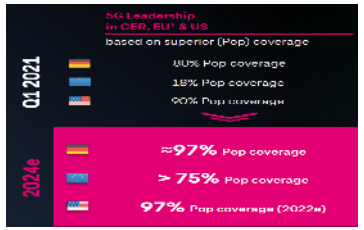
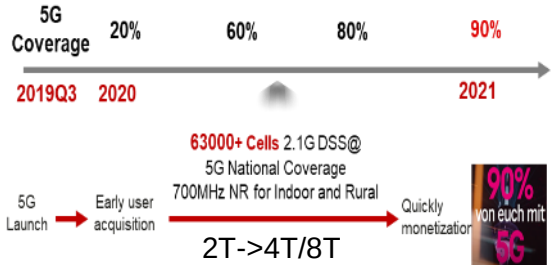
5G will bring operators to reshuffle, But experience is the KEY

Opt D：Achieve business transcendence through 5G

5G Coverage Strategy
Drive network construction



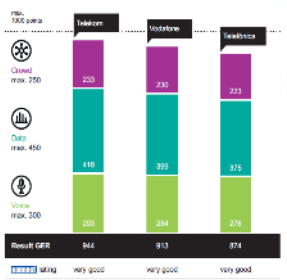
5G Coverage 20% to 90% only in 2 Years



Marketing Sharing 4G NO.2->5G NO.1

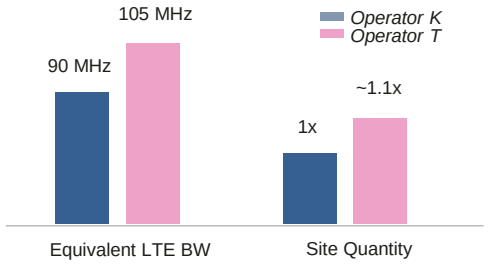
			4G	5G
Germany	DT	1	36.00%	47.42%
	VDF	2	25.54%	30.18%
	O2	3	38.46%	22.40%

P3 Keeping NO.1：Connect & Chip test



Opt K：Accelerate 4G Modernization, Overtake to be NO.1

Less Spectrum and Site Resource



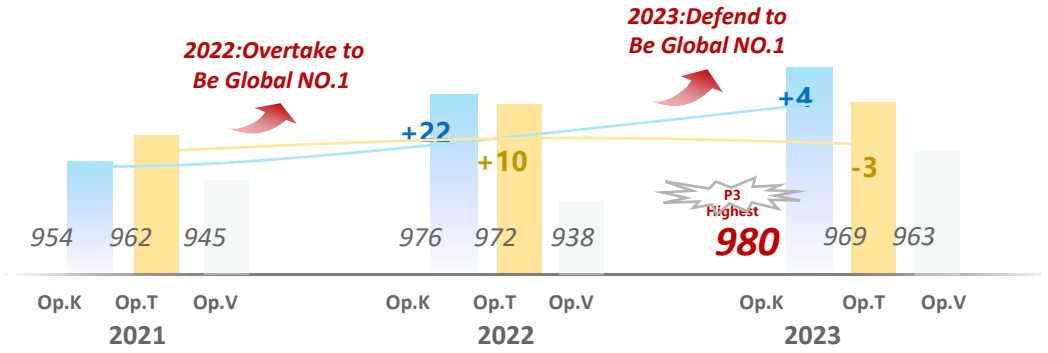
**No C-BAND in OPT K*

Multi-Antenna & Wide Bandwidth Modernization@FDD Only

- All Mid bands : 2T → **4T/8T**
- All Low bands: 2R → **4R**
- Low : Mid band site = **1:1**

LTE 3CC CA 242Mbps → **4.3x** → NR700(10M)+**LTE 4CC** EN-DC：1.0Gbps

OPT K Overtake OPT T and Get Global P3 NO.1 in 2023



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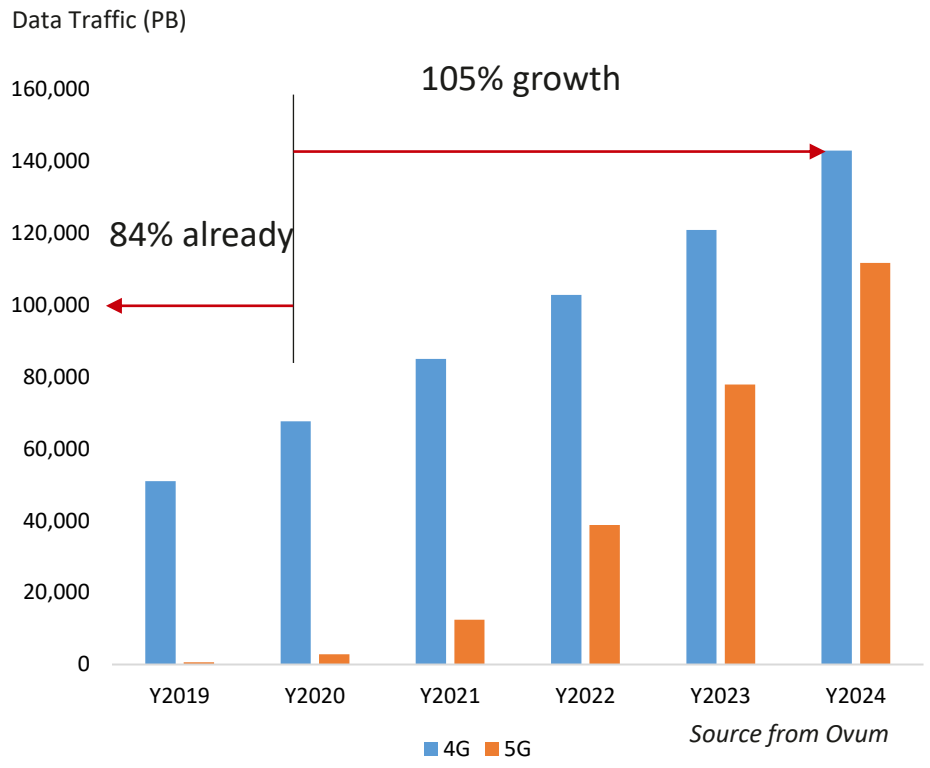
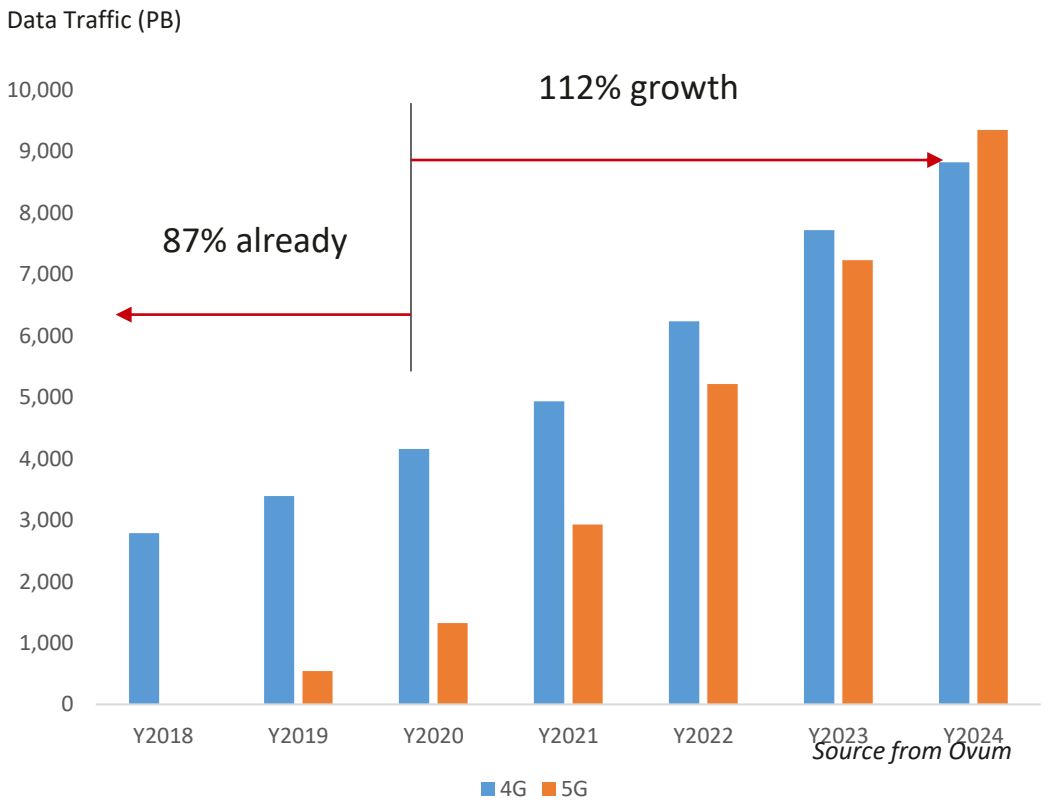
LTE Traffic Still Growing Greatly Even in China & Korea

China LTE traffic increased 84% after 5G 2 years



Korea LTE traffic increase 87% after 5G 3 years



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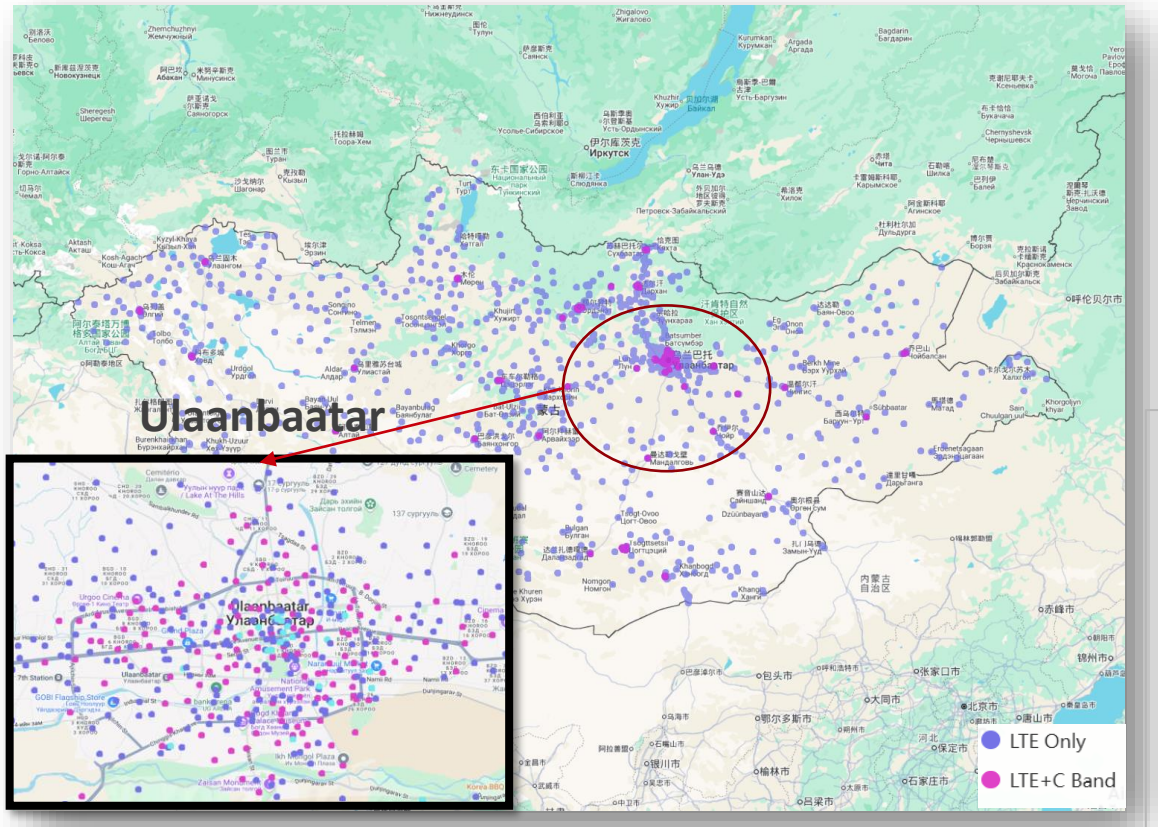
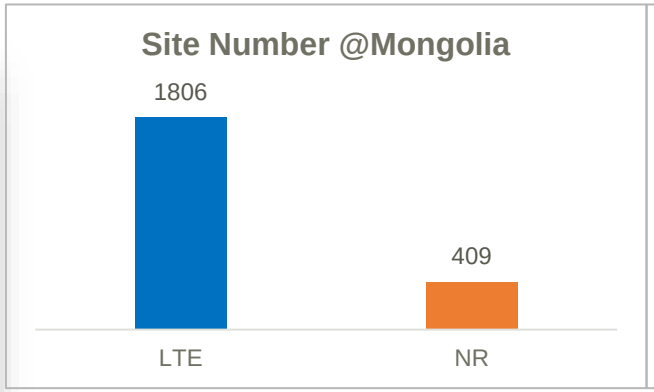
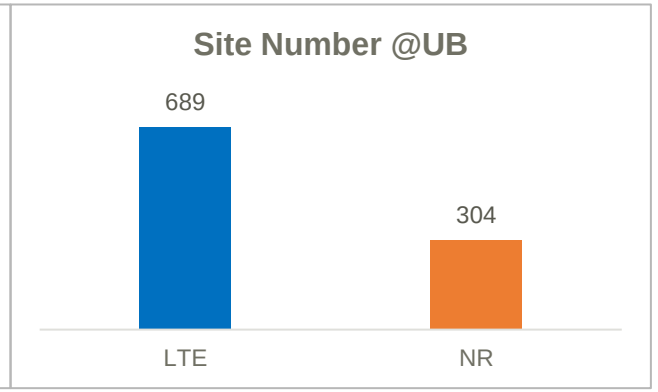
5G Mainly located in Ulaanbaatar, but 5G/4G Alignment only 44.1%

5G Site Over View

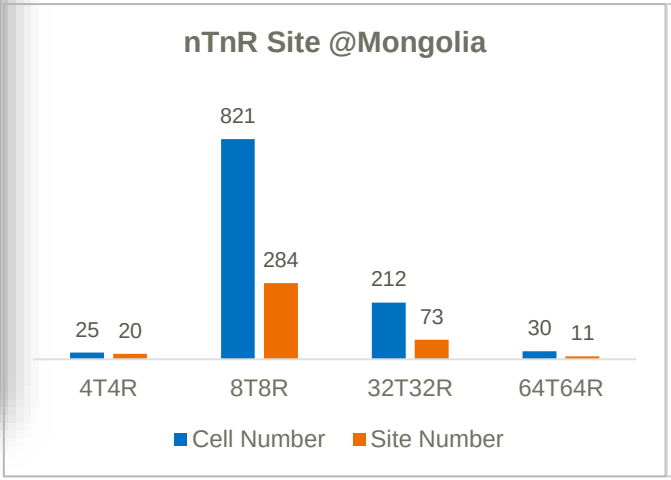


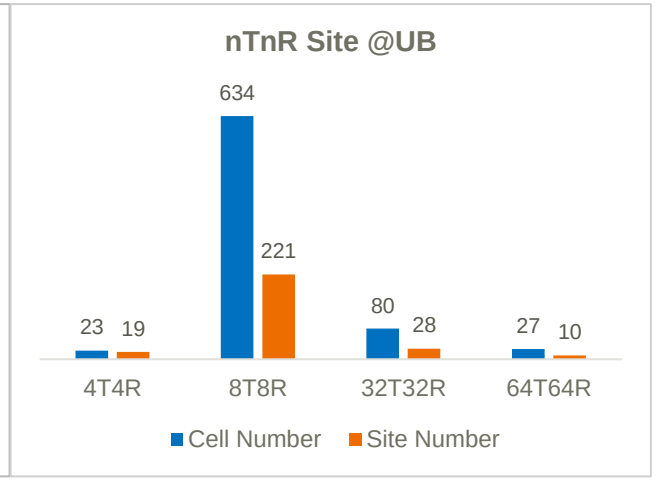
UB 5G/4G Co-site ratio is **44.1%**, 5G coverage is not continues





UB MM Site ratio is **13.6%**





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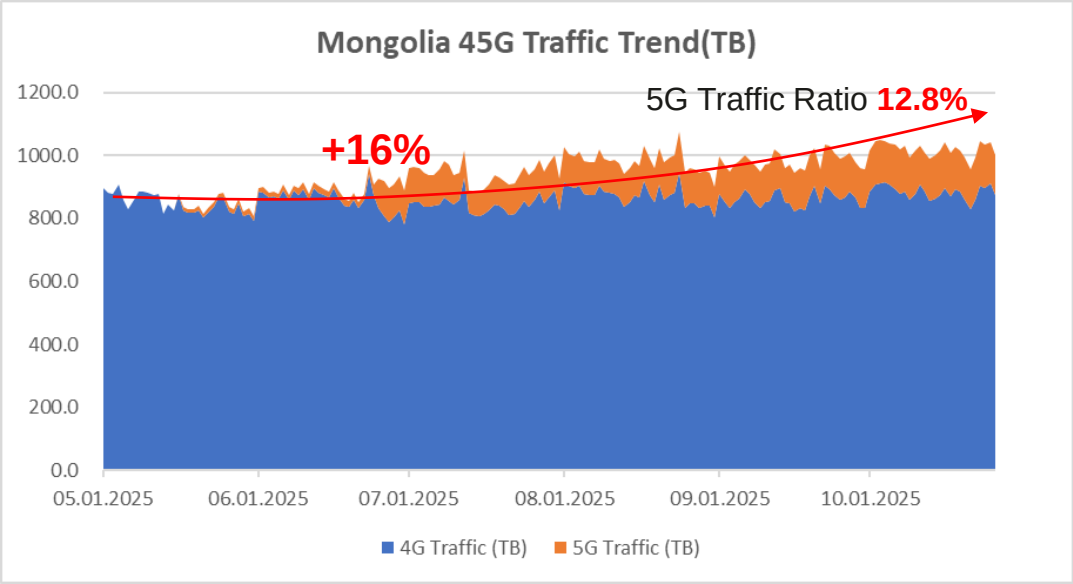
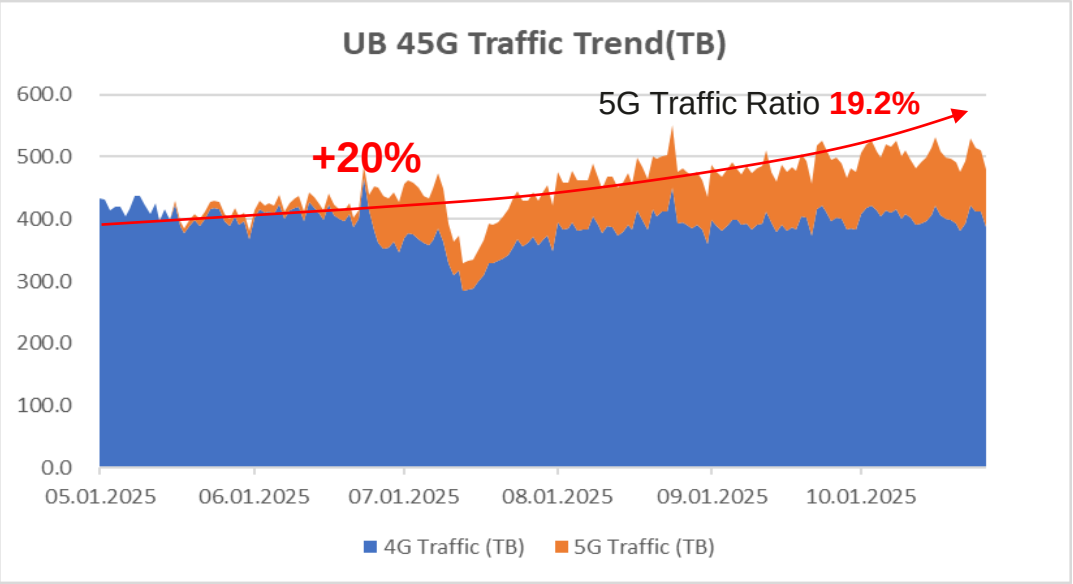
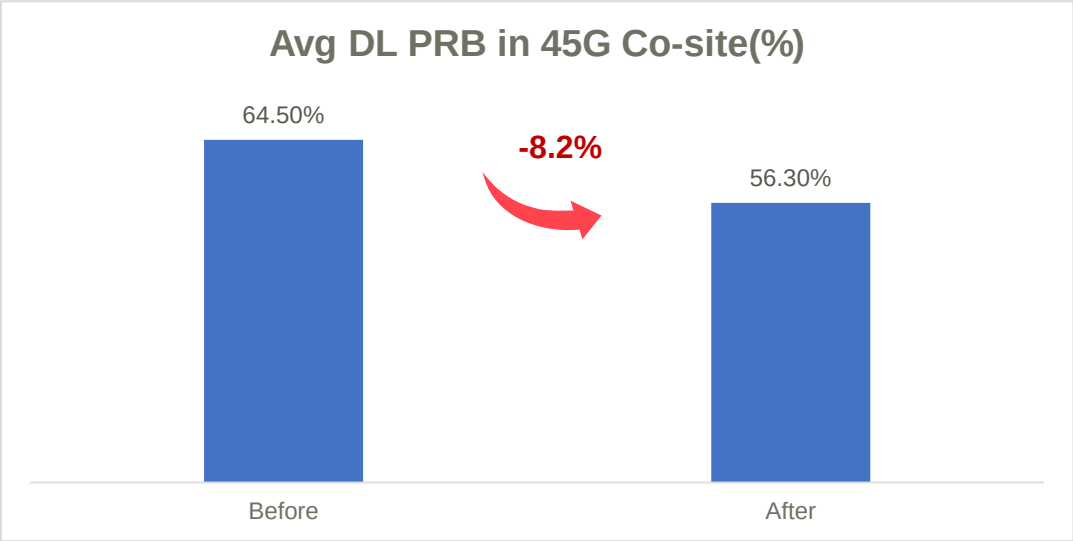
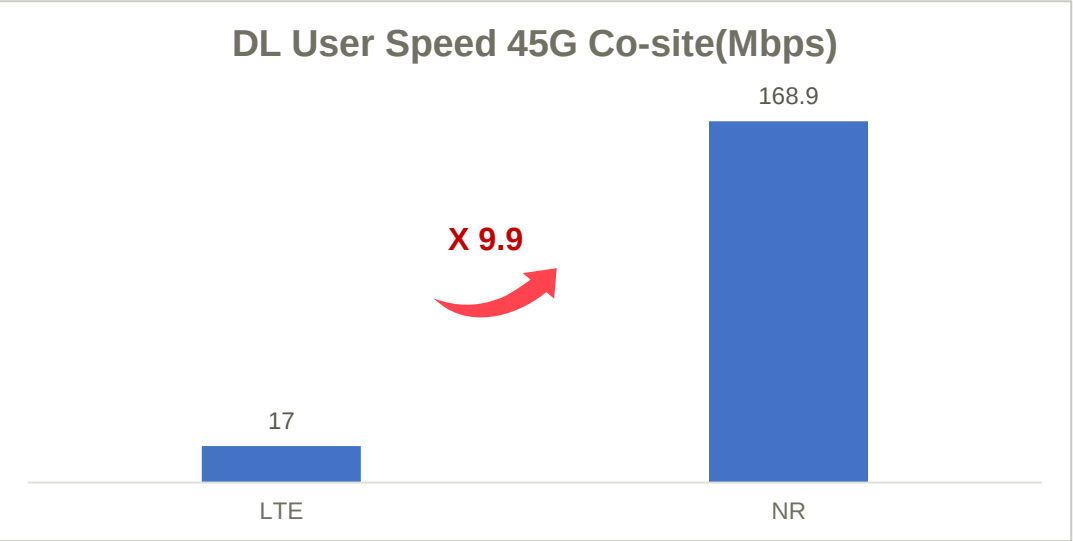
5G Traffic Ratio 12.8% and LTE Cell Load Decreased 8.2%









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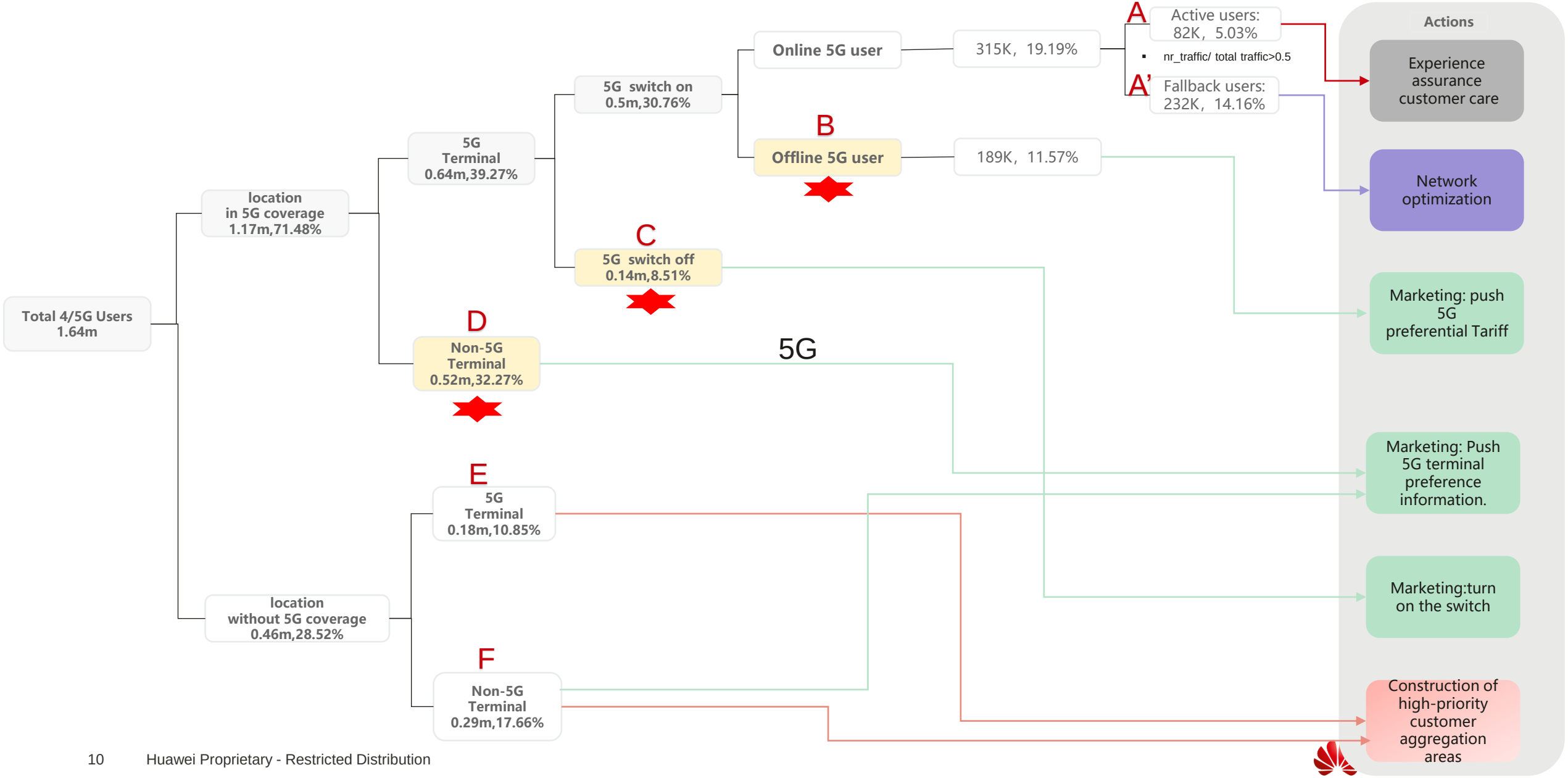
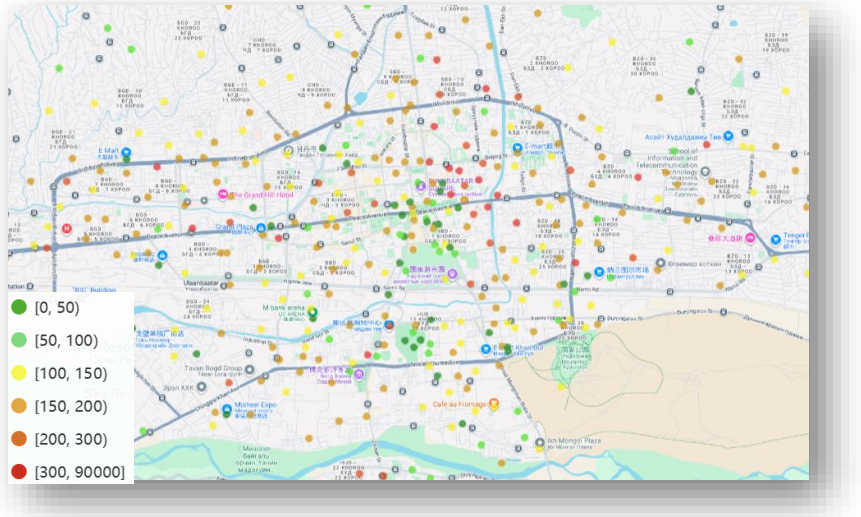
5G User Segmentation

Time Range: 2025-09-29~ 2025-10-06

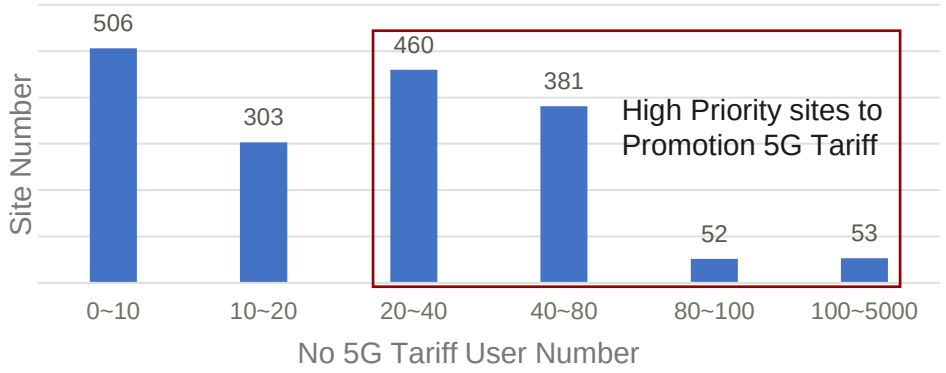


Type B: 5G Coverage area more tariff suggested to be released

Type B User Distribution(5G Terminal User)



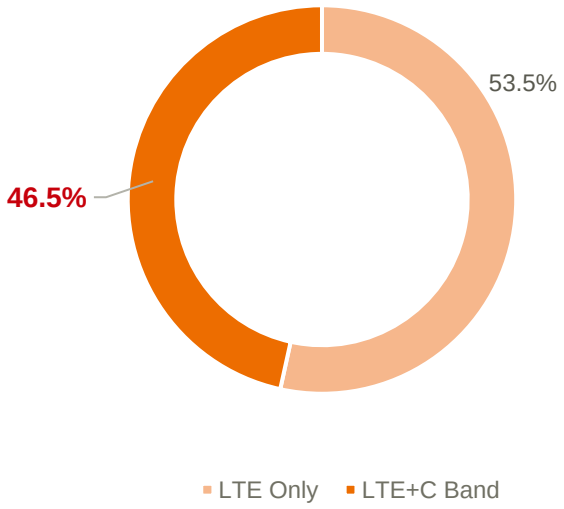
No 5G Tariff User Site Distribution



46.5% 5G no package User located in 45G Co-site

- **46.5%** 5G terminal without 5G tariff User located in 5G Site, more 5G tariff can be acquired in these area;
- **53.5%** in LTE Only Site. For these areas firstly suggested deployed 5G first;

Type B User Site Type Distribution @ UB



Type B User: *5G Terminal Switch on with no 5G-Tariff*

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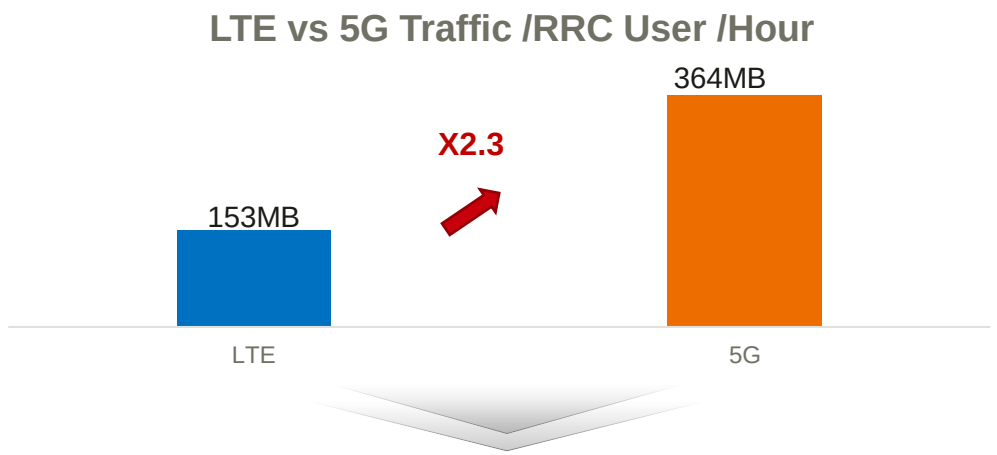
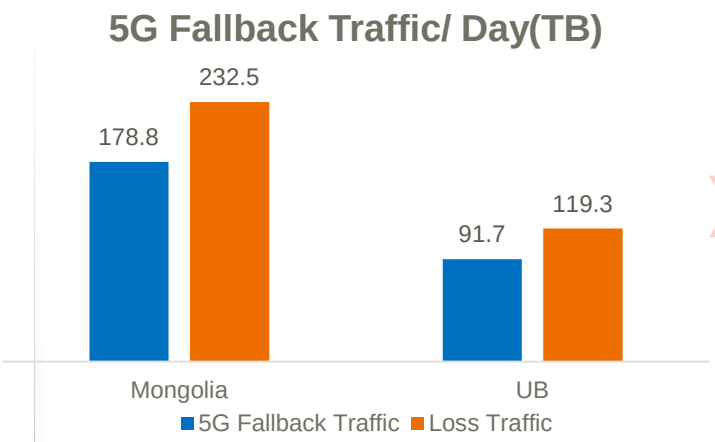
5G Discontinuous Coverage Cause 33M$ Revenue Loss

232.5TB Traffic Loss caused by 5G User Fallback





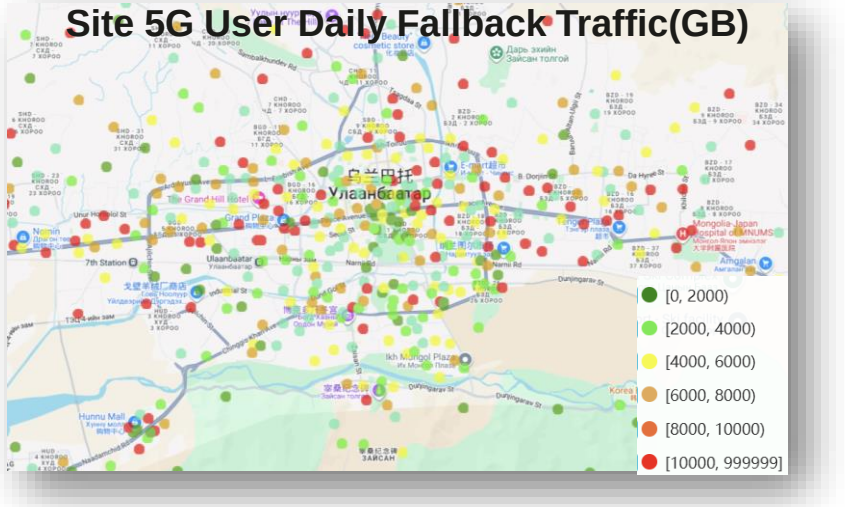
Revenue Loss

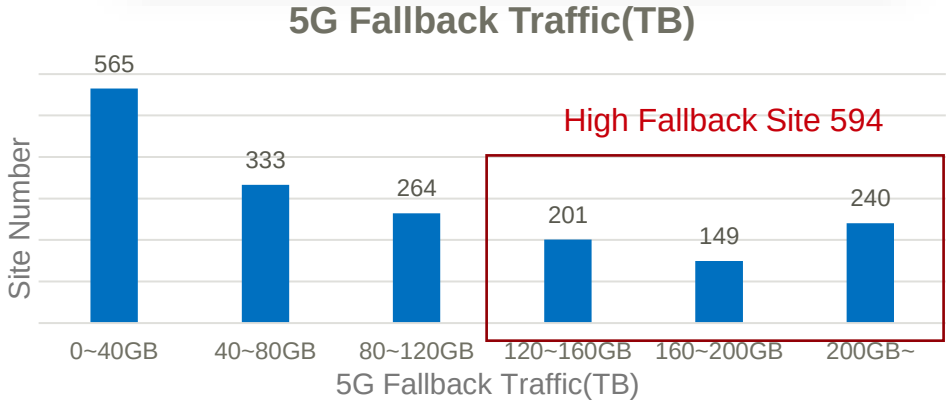
62.8k$/Day

1.9M$/Month

22.3M$/Year

594 Site 5G Fallback Traffic >120GB/Day





Note: Data Price = 0.27$/GB

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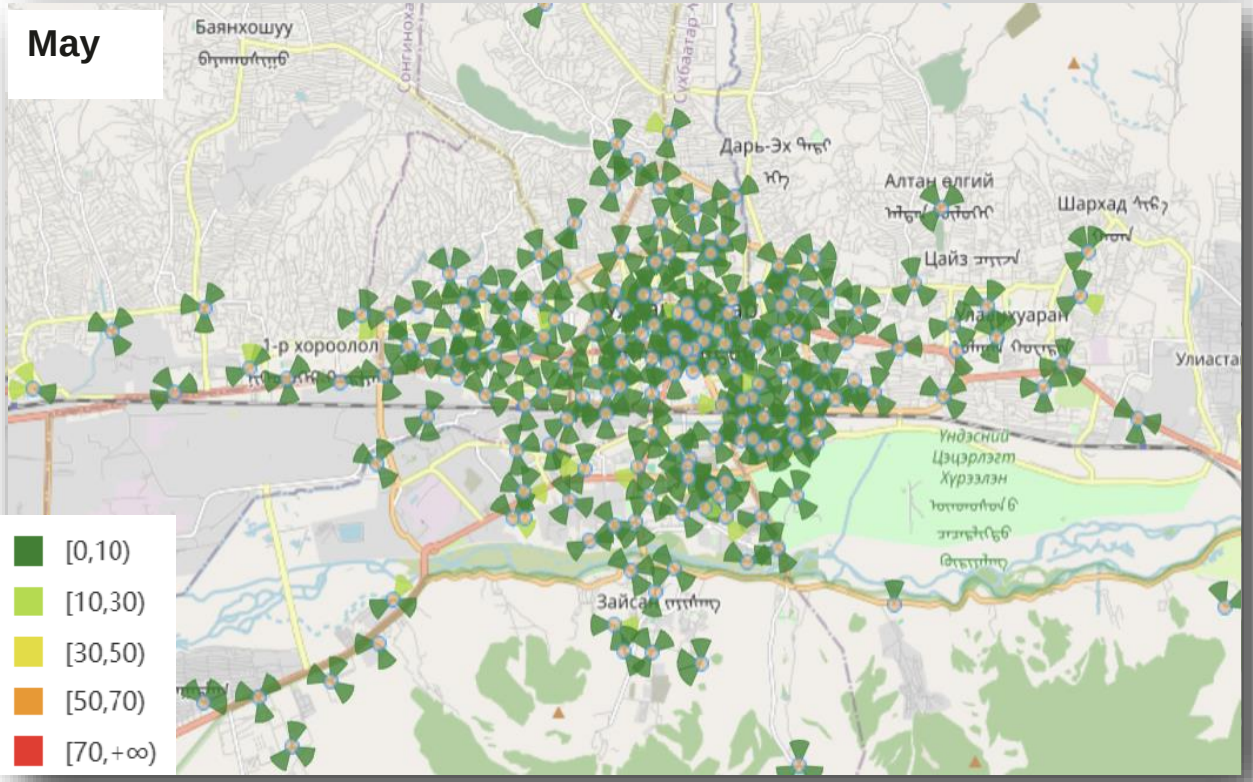
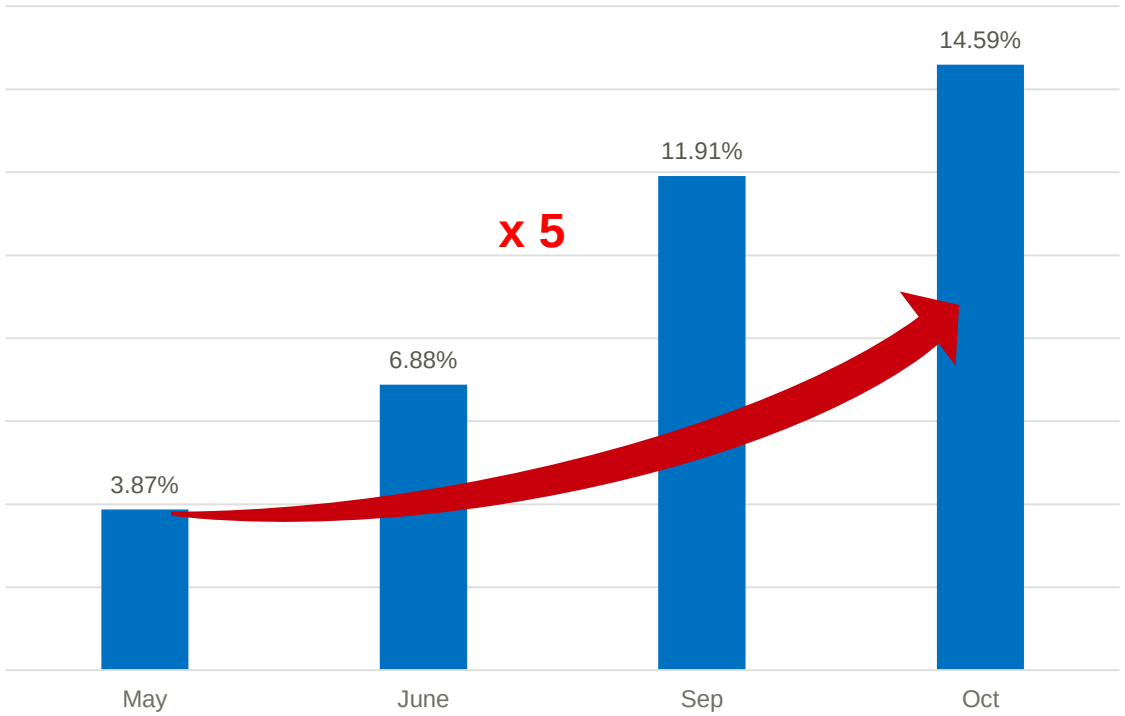
5G Cell PRB Usage increase by 5X in 5 months

5G Cell DL PRB Distribution



5G Cell Average DL PRB Usage



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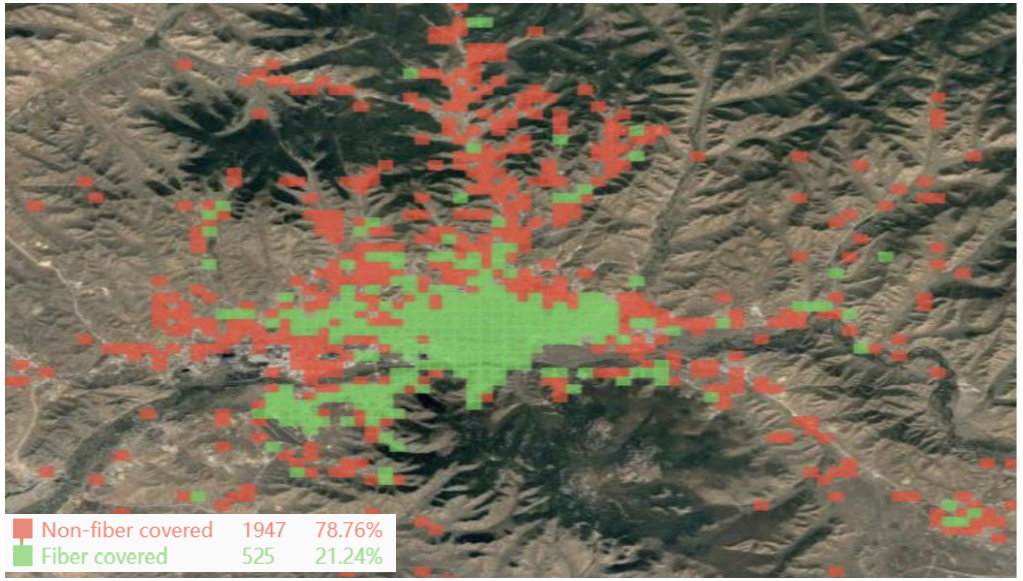
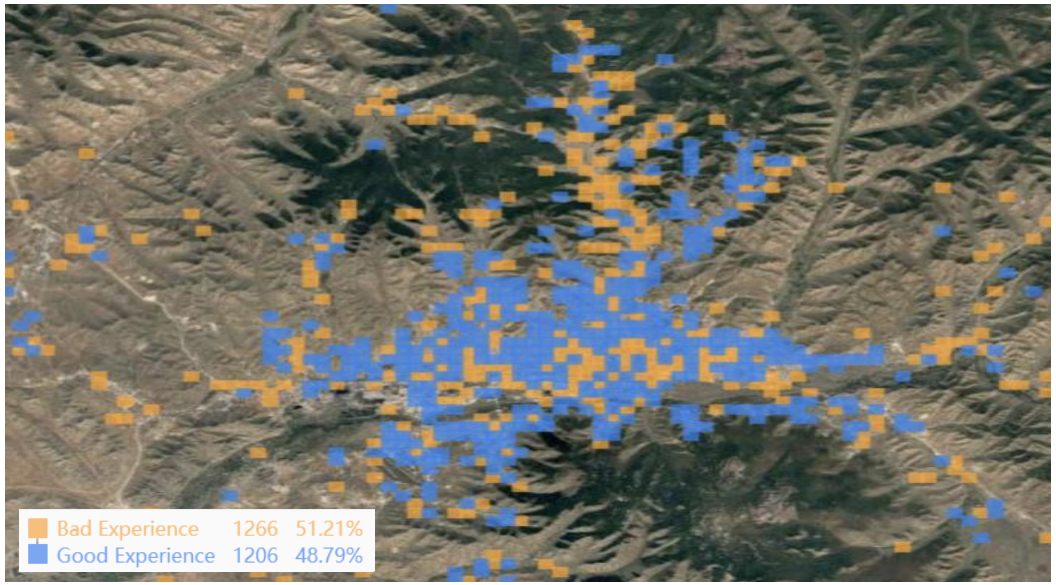
Low Experience and No Fiber area, FWA are Suggested, totally k Households

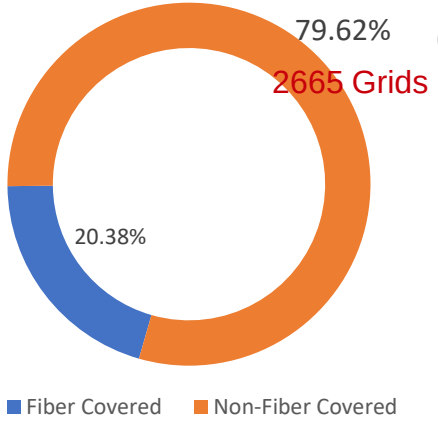
Grid Fiber Coverage



Grid User Experience



Fiber Covered Grids



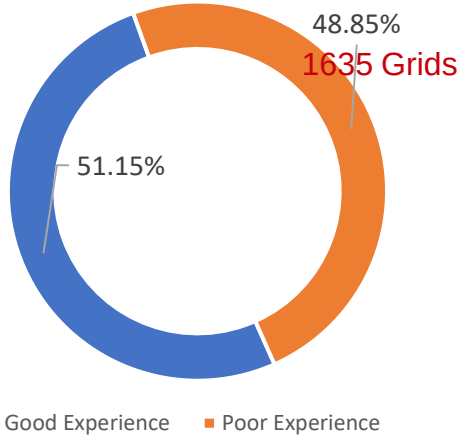
Target Area 1

(Non-Fiber + Household Density>=10)

Grid Number
1239

Households
230900

Good Experience Grids



Good Experience>50Mbps

Target Area 2

(Fiber Experience<50Mbps)

Grid Number
191

Households
39700

Grid Size: *500m*500m; Household : 5 people*

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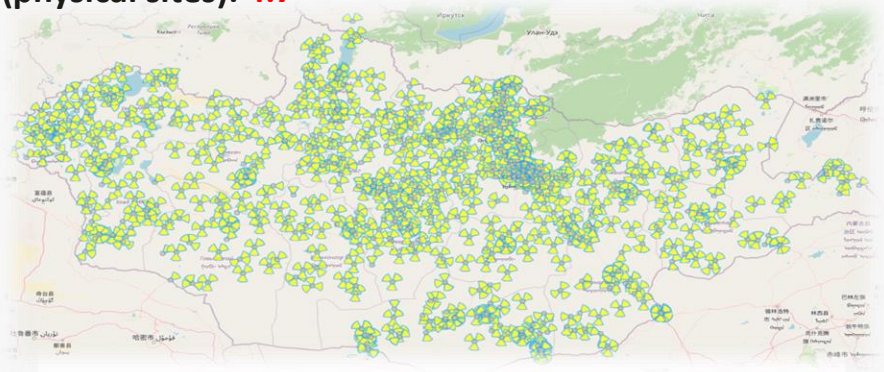
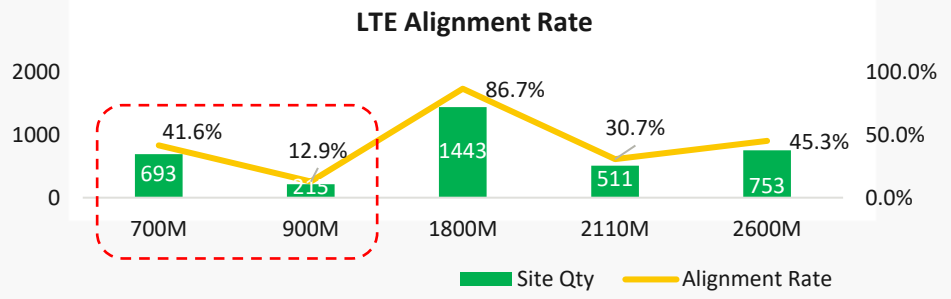
Low Band poor alignment rate and 2T base network is not conducive to evolution

Network Breadth

- 1664 LTE Sites In Total： Number of sites per 10,000 people (physical sites): 4.7



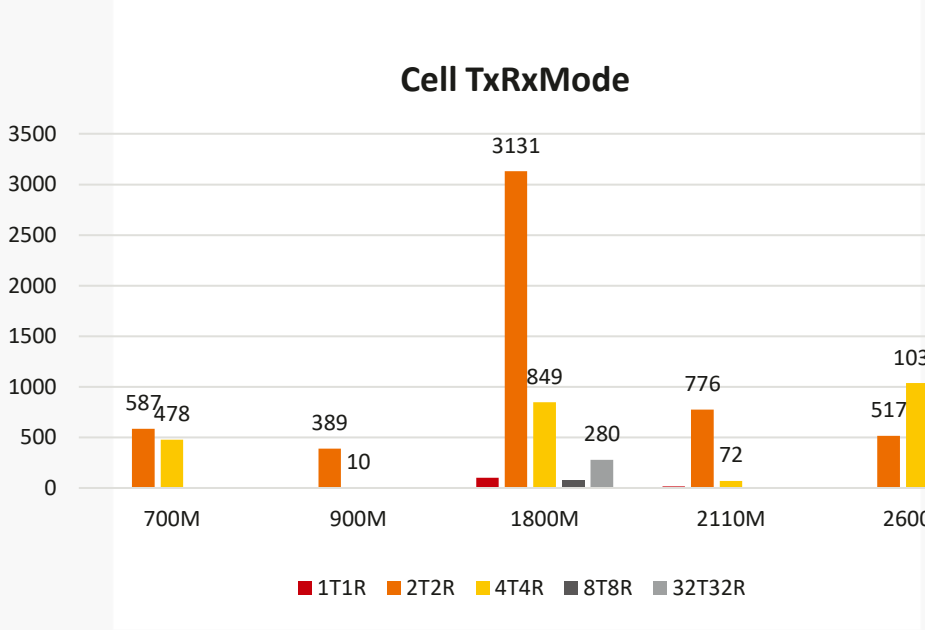
- Band alignment rate is low, and the spectrum potential is not released.



50% of sites have deployed L700/900， affecting LTE basic coverage.

Network Thickness

- The network is based on 2T, with a small amount of 8T/MM.

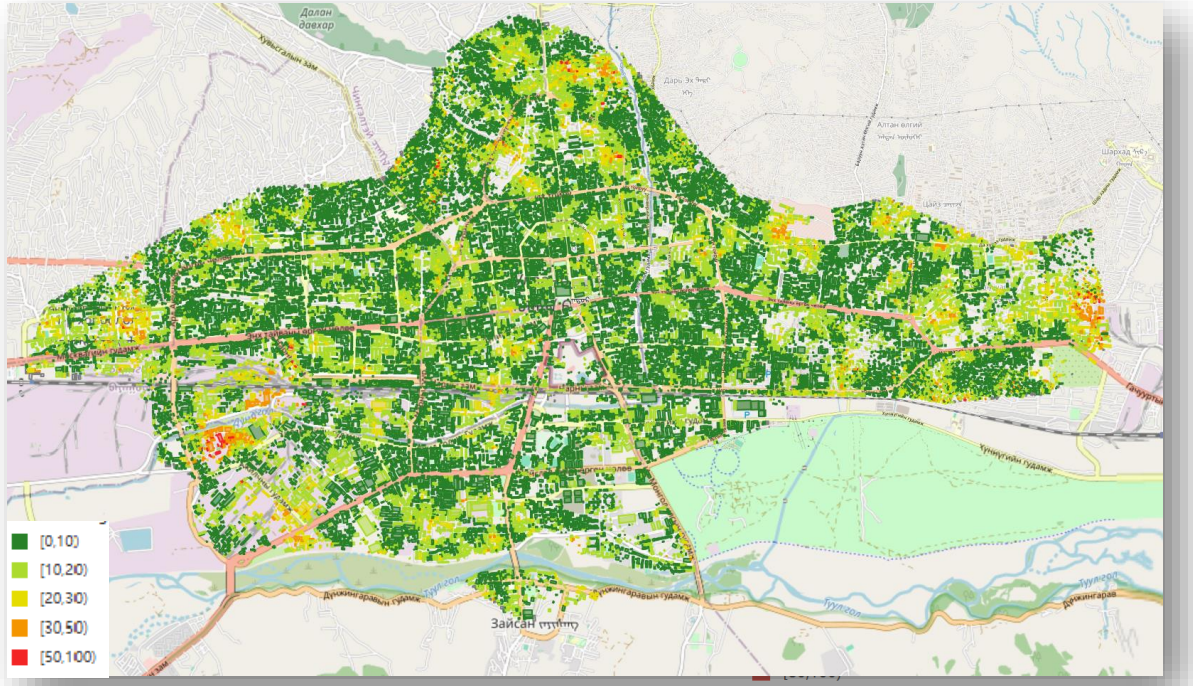


66% of cells are 2T, and the base network should evolve to 4T.

10% Buildings have poor indoor coverage

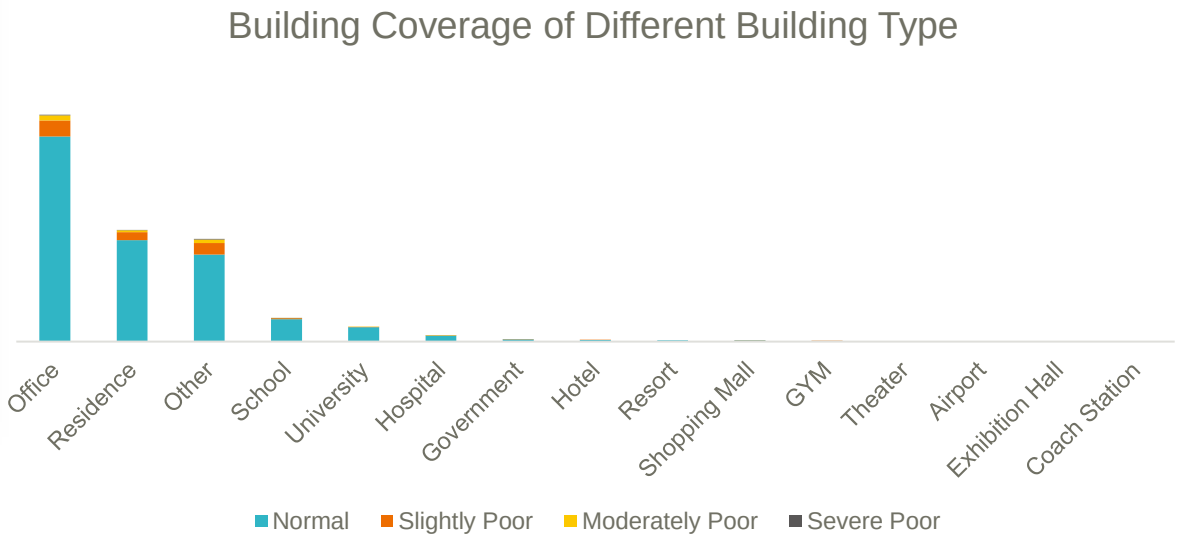
- In UB, total **34109** buildings , **3400(10%)** weak coverage buildings;
- Most weak coverage buildings located in **office** and **residence** scenarios;

Building Indoor Coverage



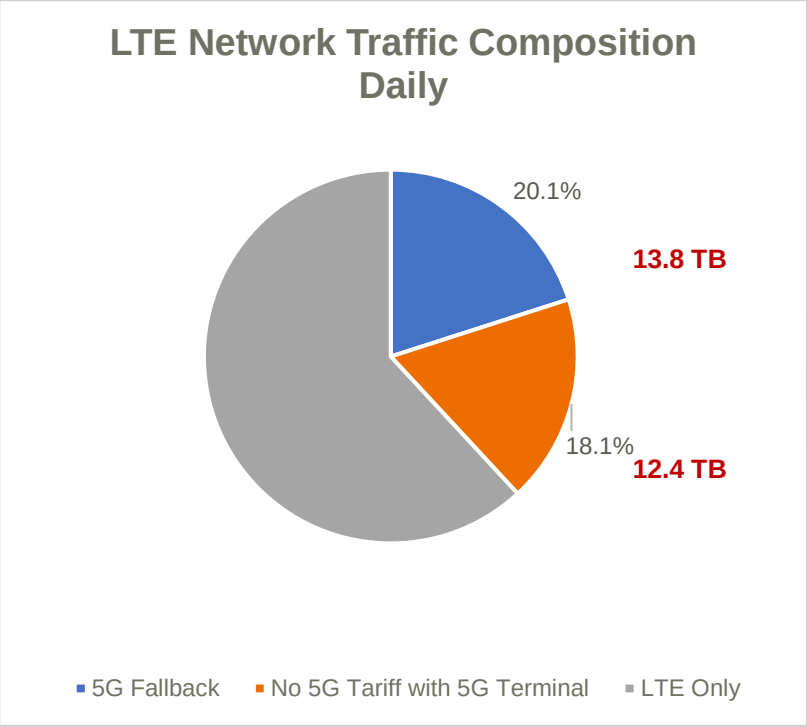
Weak Coverage Buildings In Different Scenarios

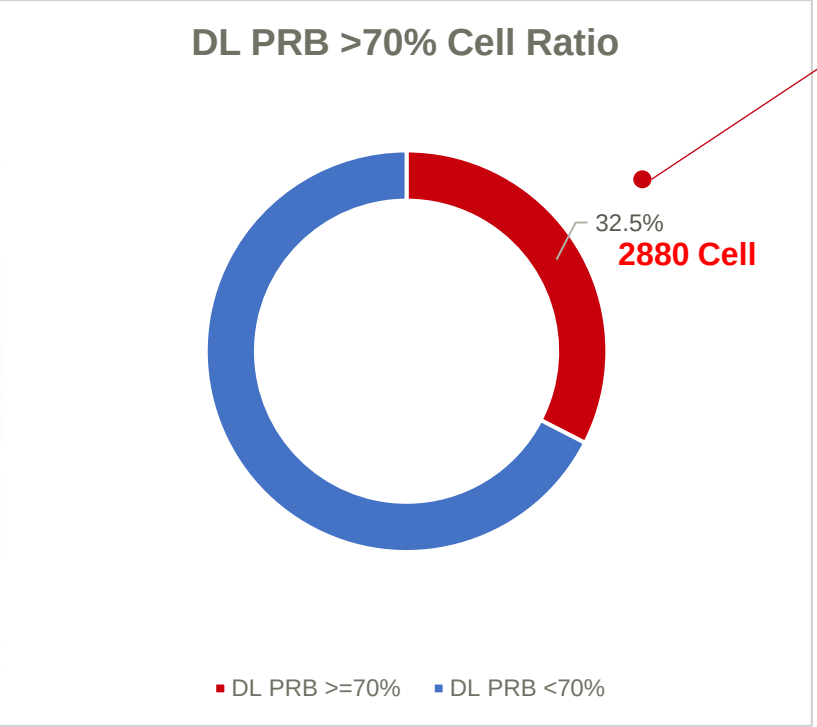
Index	Normal	Slightly Poor	Moderately Poor	Severe Poor
Weak MR Coverage Ratio(%)	<20%	20%~30%	30%~50%	50%~100%
Building Number	30467	2570	740	90

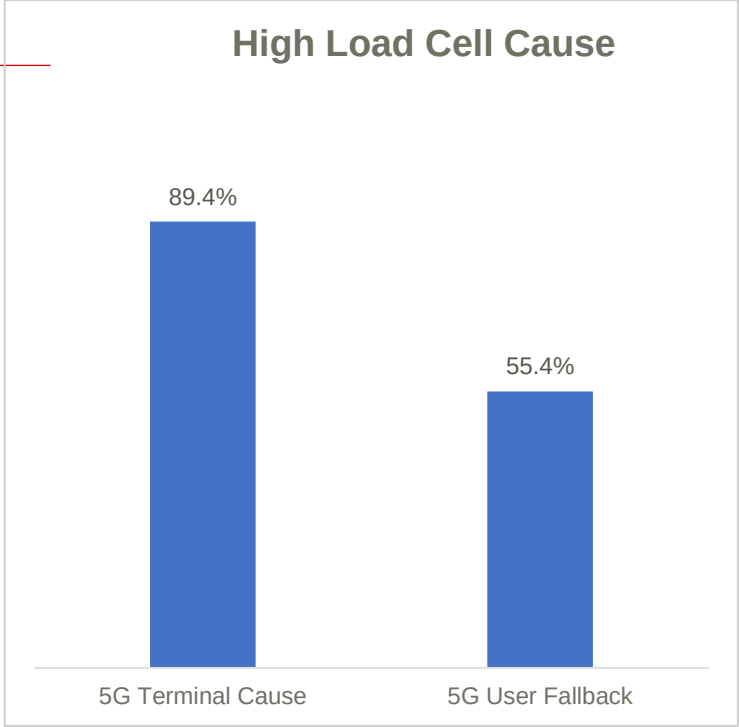


Building Coverage: *Building MR <-105dBm MR Num/ Building Total MR*
Volte MOS>3.5 LTE coverage should reach >-105dBm

5G User/Terminal in LTE Network Cause More High Load LTE Cells







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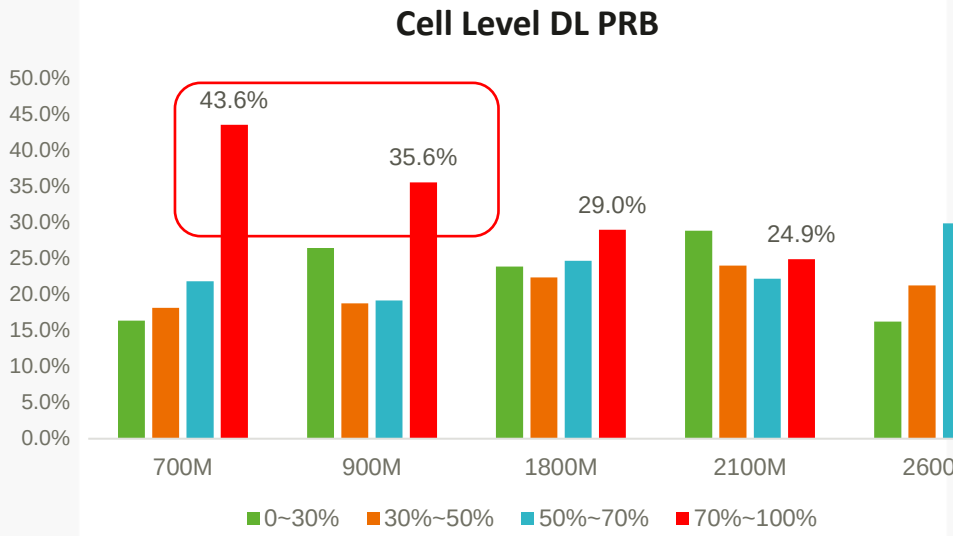
31% of Cell PRB>70%, 22% Cell 5Mbps Experience Satisfaction<60%

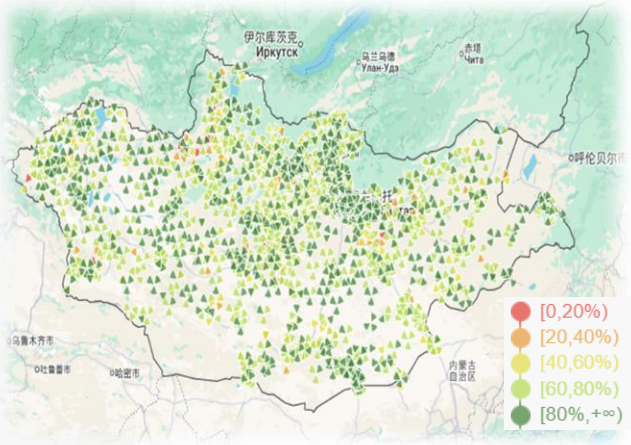
Network Load

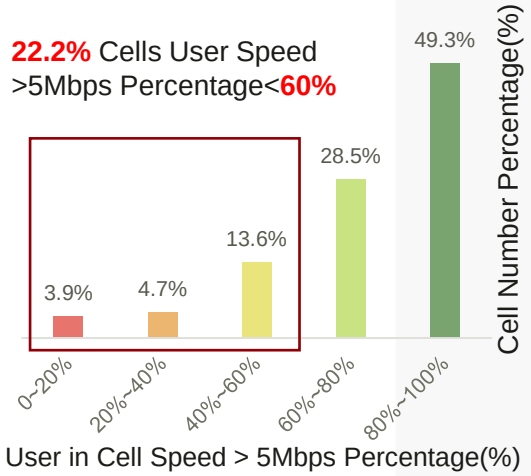


- **At least 31.6% of the cells have PRB > 70%, especially in low band**

Network Experience

Cell 5Mbps Experience Satisfaction Percentage





- **22.2% of Cell 5Mbps Experience Satisfaction <60%, affecting user experience.**

Overloaded sites cause traffic suppression and poor user experience. Expansion is required.

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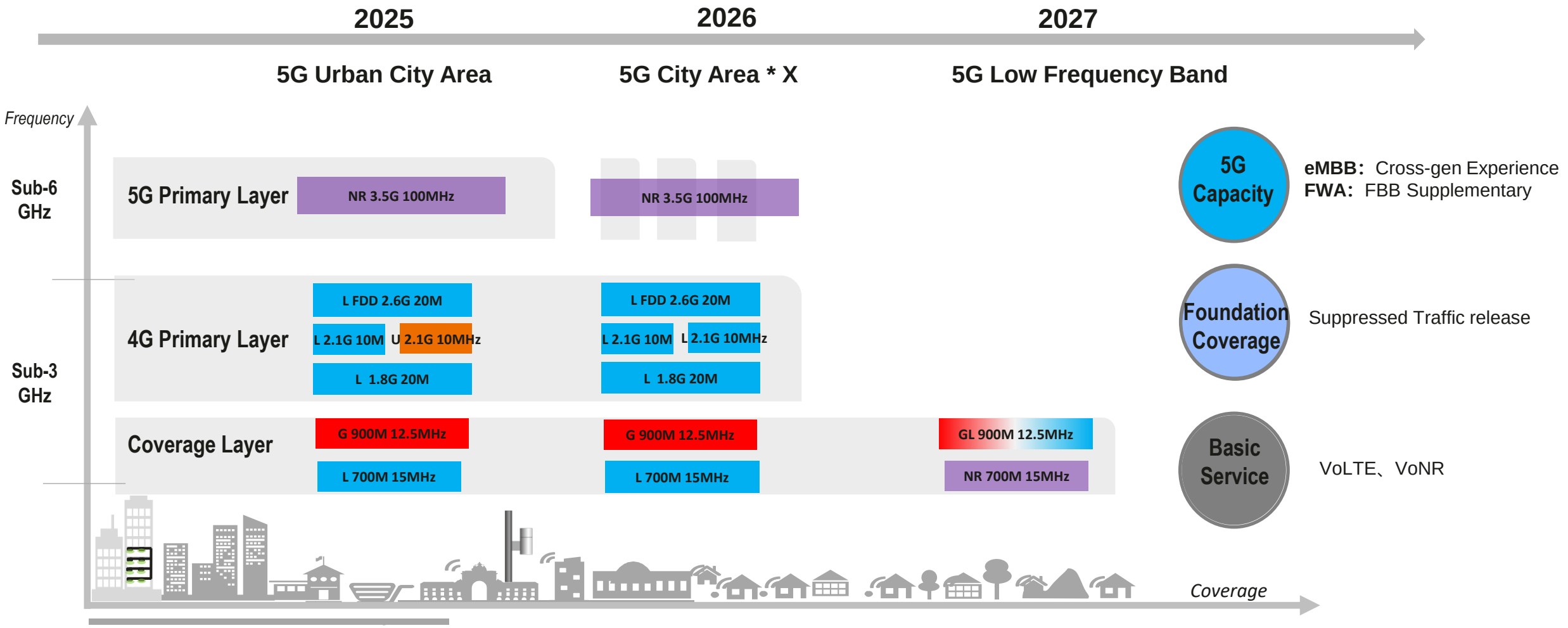
Contents

- Industry Trend
- 5G Network Insight
- 4G Network Insight
- Network Planning Suggestion

Future Oriented Target network: Diversified Solution for All-Scenarios



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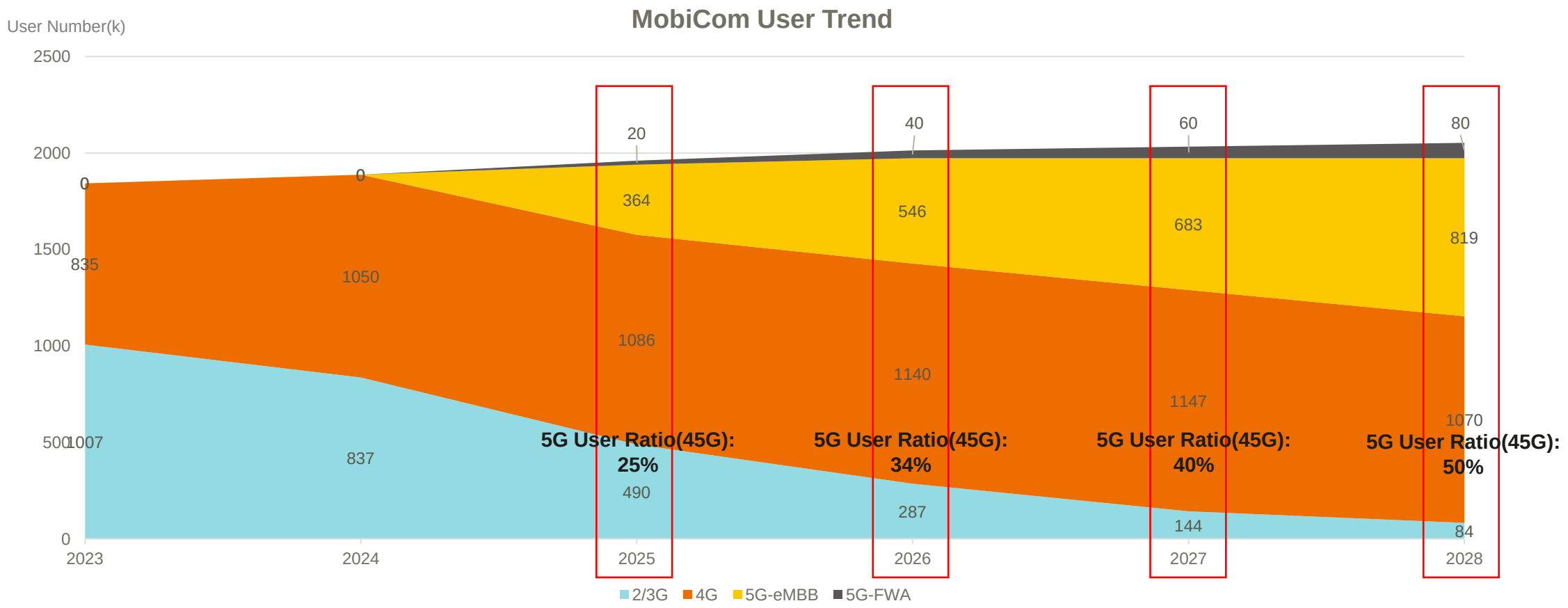
Until 2027, 5G User will Occupy 40% of Total 45G User



Data Source: GSMA + Other Operator Exp.

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By 2028 5G traffic will reach 50.3% and LTE will reach peak in 2027

4/5G Coordinated Traffic Forecast

Natural Growth x Dou Increase(1) x 5G eMBB User x 5G Camping Ratio

Traditional 4G Forecast Model

HD Video, use habit

= 5G Traffic

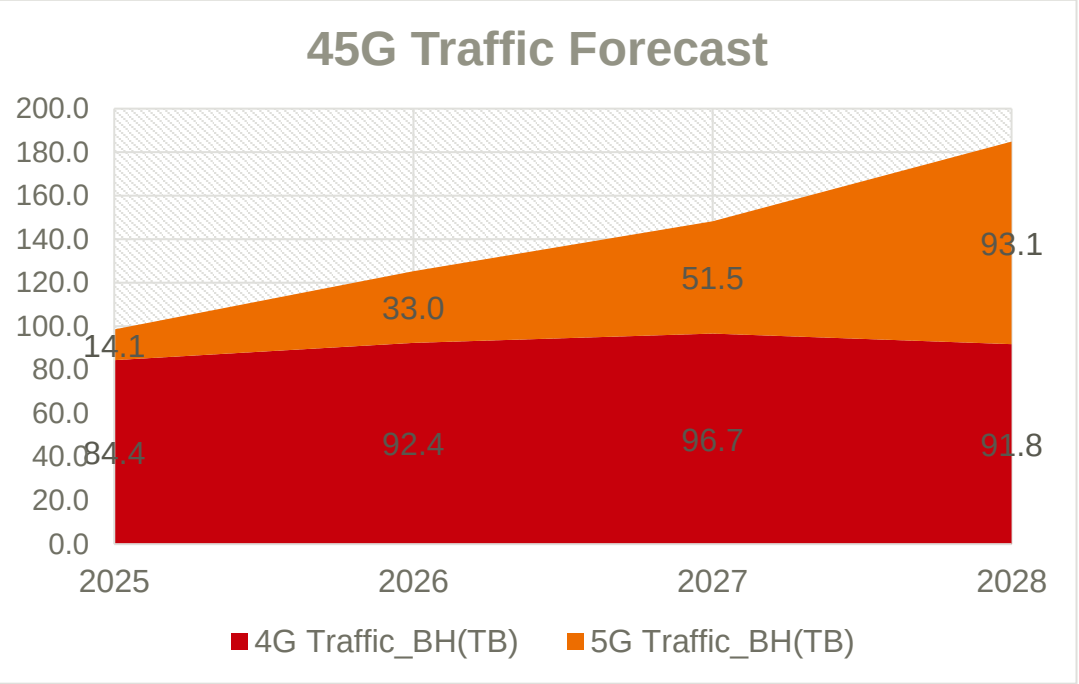
+ 5G FWA Dou x 5G FWA User x 5G Camping Ratio

x(1 - 5G Camping Ratio)

Natural Growth x 4G Penetration increase Factor + 5G Fallback Traffic = **4G Traffic**

1 -5G Penetration-2/3G

By 2028, 5G Traffic will occupied by 50.3%



	2025	2026	2027	2028
5G Traffic Ratio	14.3%	30.5%	41.2%	50.3%
5G Camping Ratio	43.5%	62.0%	77.8%	92.0%

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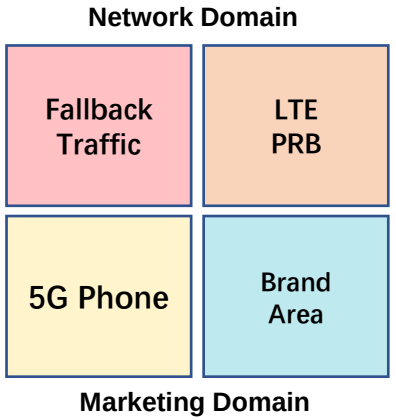
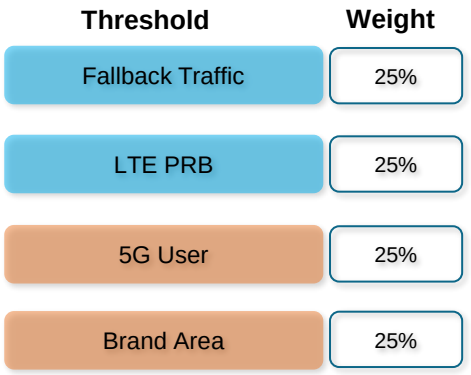
3.5G Coverage Planning: NR Planning Method for One-Deploy Multi-Benefits

① Multi-dimension Insight

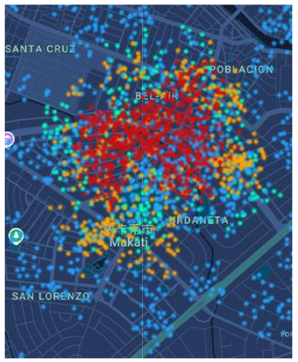


② Weight Scoring



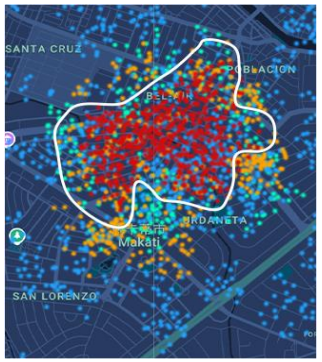
③ Value Site Selection

Site ranking by score



④ HVA Aggregation

Cont. HVA by density

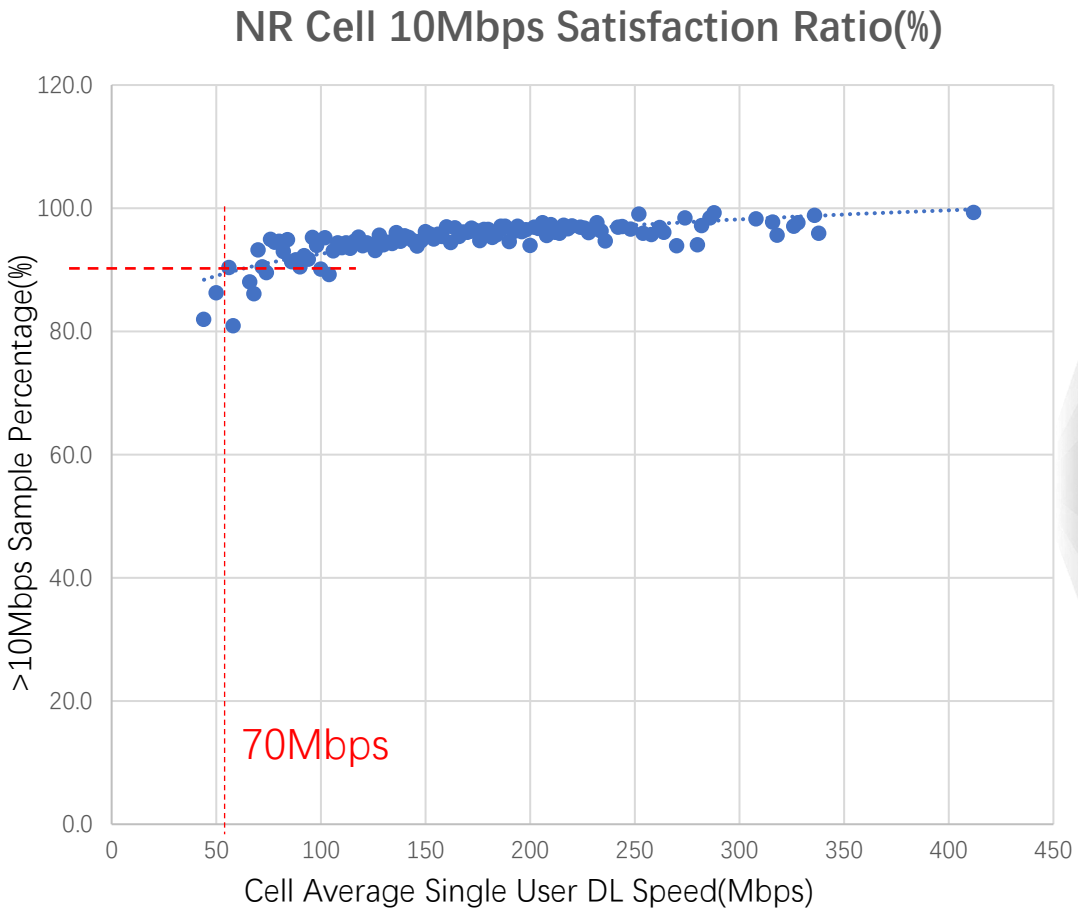


Region	2026 Score (>=80)	2027 Score (60~80)	2028 Score (50~60)	2029 Score (40~50)	2030 Score (20~40)
Total	397	288	207	204	177

Planning Result

5G Capacity Expansion(Sector Level)-eMBB

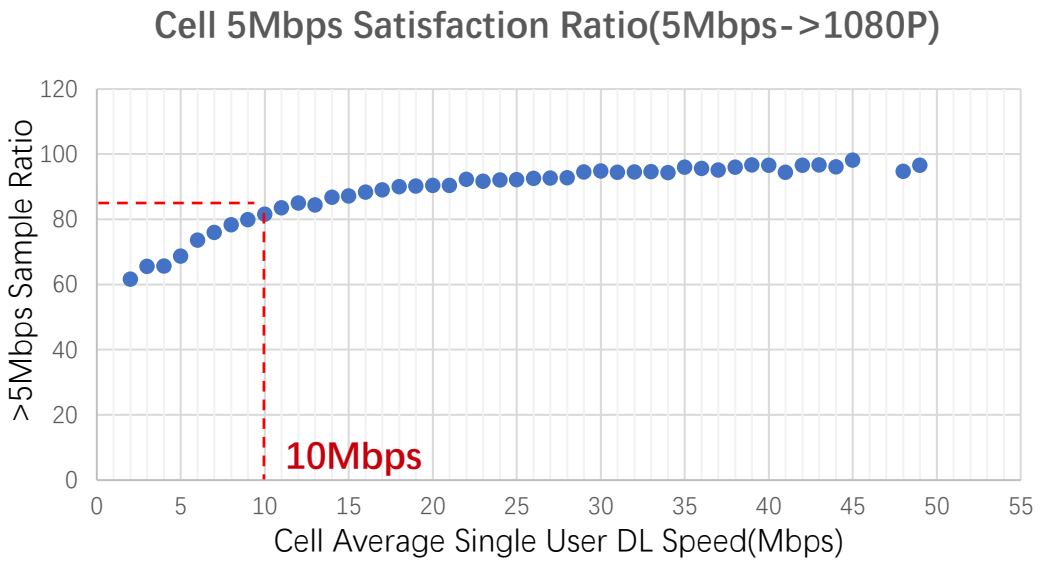
5G Capacity Expansion Criterial: 10Mbps@90% -> Cell Average User Speed **70Mbps**

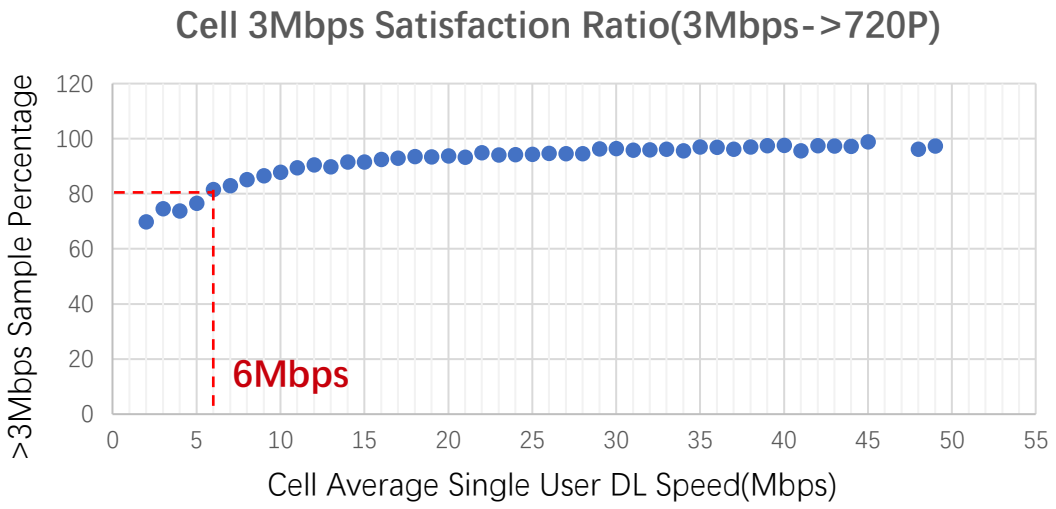


Original	Solutions	2025	2026	2027	2028
8T8R	8T8R->32T32R	22	48	134	482
	8T8R->32T32R+Feature	1	1	5	10
	8T8R->64T64R	0	1	1	2
	8T8R->64T64R+Feature			1	5
	8T8R->MetaAAU	0	0	0	
	8T8R->MetaAAU+Feature				2
32T32R	32T32R->Features	3	4	14	41
	32T32R->64T64R	0	2		
	32T32R->64T64R+Feature			3	15
	32T32R->MetaAAU	0	2		
	32T32R->MetaAAU+Feature	1		4	5
	32T32R->New Site				5
64T	64T64R->Features	2	4	4	4
	64T64R->MetaAAU	1	3		
	64T64R->MetaAAU+Feature			2	1
	64T64R->New Site			3	8
SUM		30	**65**	171	580

4G Capacity Expansion(Sector Level)

4G Capacity Expansion Criterial: **UB:** **5Mbps@80%** -> Cell Average Speed **10Mbps**
Non-UB: **3Mbps@80%** -> Cell Average Speed **6Mbps**

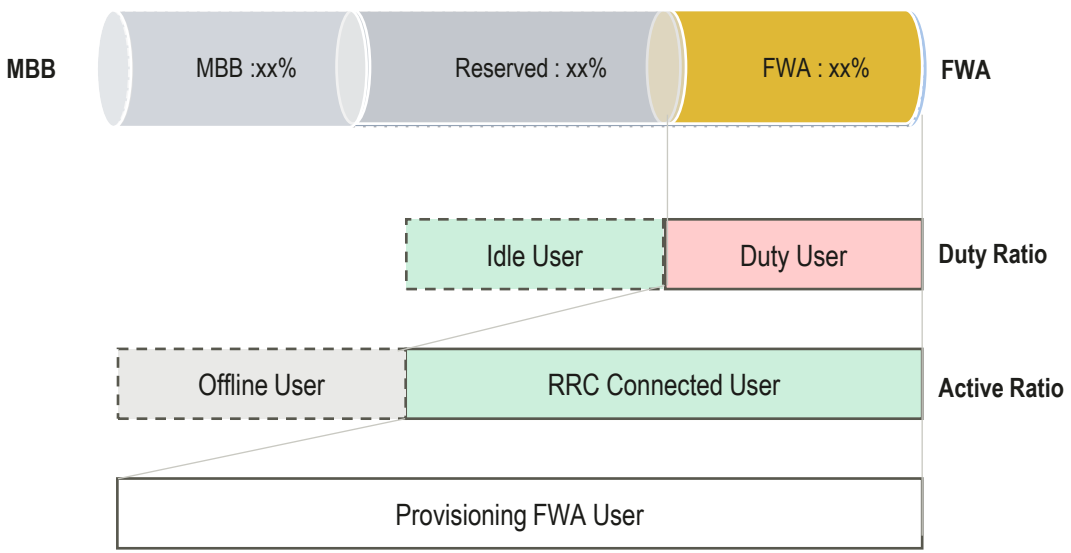




Region	Solution	2025	2026	2027	2028
UB	700M 4T4R	13	14	7	1
	1800M 4T4R	52	65	56	
	1800M 8T8R	4	9	2	
	1800M MM			6	1
	2100M 4T4R	89	82	63	3
	2100M 8T8R		3		
	2600M 4T4R	2	15	14	1
	2600M MM	12	12	13	5
	1800M+2100M 4T4R	22	22	7	
	1800M+2100M 8T8R	49	34	50	4
	1800M+2100M MM	9	13	19	11
	1800+2100M+2600M MM	1			1
NonUB	700M 4T4R	125	173	205	27
	1800M 4T4R	85	92	83	328
	1800M 8T8R	1			
	1800M MM	2		3	1
	2100M 4T4R	21	37	101	59
	2600M 4T4R		4	1	1
	2600M MM	3	2	2	
	1800M+2100M 4T4R	3	4	3	3
	1800M+2100M 8T8R	1	6	11	8
	1800M+2100M MM		2	4	3
	1800M+2100M+2600M 4T4R		1		
	1800M+2100M+2600M MM	1			
	SUM	495	**590**	650	457

FWA Users/Site : 100~300Mbps, 100 Users/Site in world wide

Provisioning FWA User Formula



$$\text{Provisioning FWA User Number} = \frac{\text{Site Capacity * FWA PRB Usage Ratio}}{\text{FWA User Speed * Duty Ratio * Active Ratio}}$$

Global Ref. : DM > 200M, EM ~ 100M; PRB ~ 30%

Region	Country	Operator	Spectrum (2.3/2.6/C-Band)	Avg User DL Speed/Cell (Mbps)	5G PRB	FWA User/Site
Middle East	Saudi Arabia	Z	2.6G：90M 3.5G：100M	257	34%	115
Middle East	Saudi Arabia	S	2.3G：80M 3.5G：200M	249	26%	120
Middle East	Kuwait	Z	2.6G：100M 3.5G：100M	181	41%	70
Middle East	Oman	O	3.5G：100M	113	26%	60
Middle East	UAE	D	2.6G：100M 3.5G：200M	234	11%	60
Africa	South Africa	R	3.5G:140M	147	53%	160
Asia	Philippines	D	3.5G:140M	103	12%	30

To reach FWA User Number target, more NR resources are needed

Prediction of FWA User allocate per Site

NR Site Config	3.5G 100MHz * 8T8R * 3 Sector			3.5G 100MHz * 32T32R * 3 Sector			3.5G 100MHz * 64T64R * 3 Sector			3.5G 100MHz* MetaAAU 64T64R*3 Sector		
Site Capacity(Mbps)	753			1767			2295			2983		
eMBB Resource	30%											
Reserved	20%											
FWA Resource	50%											
FWA Speed-based Tariff	60Mbps	100Mbps	200Mbps	60Mbps	100Mbps	200Mbps	60Mbps	100Mbps	200Mbps	60Mbps	100Mbps	200Mbps
User Number to Allocate	30	18	9	75	45	21	96	57	30	125	74	39
Package Ratio(%)	50%	30%	20%	50%	30%	20%	50%	30%	20%	50%	30%	20%
FWA User Number mix	23			56			72			93		

CPE：4T4R

Live Network Capacity for FWA User

MM	8T8R	32T32R	64T64R
Capacity	23	56	72
Live Net Site Number	X 304	X 73	X 14

FWA User Capacity: **12088**

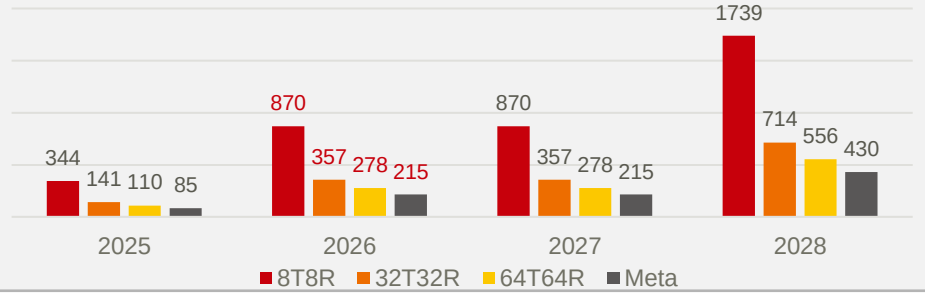
VS

Target	FWA User Number
2025	20000
2026	40000
2027	60000
2028	100000

GAP

8T or 32T or 64T

Site Number for FWA User



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FWA Easy Provision： Potential Area Identification and Precise Speed Estimation by WTTx Suite

Find Users

Identify package provisioning areas

Region selection from passive sales to active identification

HVA Identification

- ✓ Identification of HVA@ 80%
- ✓ Regional CPE allocation space estimation

Package Design for Users

Multi-dimensional experience package provisioning

Provisioning from single to multi-dimensional experience

Provisioning based on coverage/throughput/delay

- ✓ Service provisioning assurance @ 70%
- ✓ FWA/eMBB collaborative allocation accuracy @ 90%

By WTTx Suite

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Thank you.

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每个组织，构建万物互联的智能世界。

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Target Rate Setting : based on live network and future service characteristics

5G Intergenerational Experience: No Frame Freezing -> Click and Start

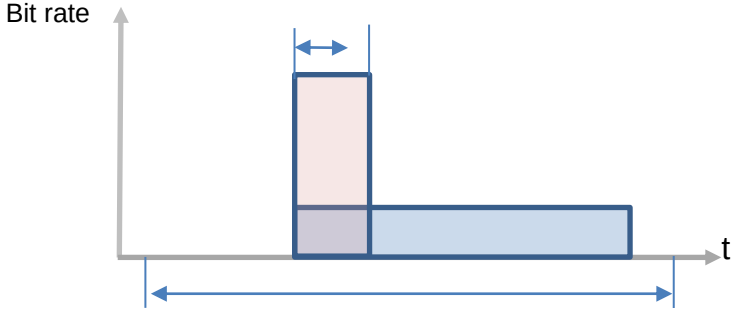
5 to 10 times the bit rate on point-to-point

To download content within 0.5 to 1s, download content within 5s, that is 5 to 10 times the bit rate.

The player initially plays the content that is buffered for 5 seconds to avoid frame freezing.

Quick response by dragging requires 6 times the bit rate

Download the content within 0.3s, that is, 6 times the bit rate.



Play the content buffered for 2 seconds after the player drags the content.

5G experience requirement analysis: the mainstream rate required currently being 10 Mbit/s to 20 Mbit/s

Typical Application	OTT	Resolution	Video Bit Rate (Mbit/s) (even playing)	4G Without Frame Freezing Required User-Perceived Rate (Mbit/s) (no frame freezing)	5G Instant Response/Click-and-Start Required User-Perceived Rate (Mbit/s) (no loading and no frame freezing)
Long video	iQIYI, Youku, and Tencent, Youtube	480p	0.6	1.2	3
		720p	1	2	5
		1080p	**2**	**4**	**10**
		1080p HDR	4	8	20
Short video	TikTok, Kwai, and Bilibili	576p	0.9	1.8	4
		720p	1.1	2.2	5
		1080p	**2**	**4**	**10**
	Facebook	360P	0.6	1.2	3
		480P	1.5	3	7.5
		720P	2.4	4.8	12
		1080P	**4**	**8**	**20**
Live audio/video	Live streaming, voice chat, video conferencing, and distance education	480p	0.8	2.4	
		720p	2	6	
		1080p	**4**	**8 to 12**	
Free-viewpoint	Youku, Migu, Wo Video, and Kugou	1080p	15	22.5	70
Cloud gaming	Tencent START, NetEase cloud games, etc.	720p	5	7.5	
		1080p	10	15	

At the early stage of 5G networks, a user-perceived rate of 10 Mbit/s to 20 Mbit/s is required to ensure experience continuity from 4G to 5G. With future service development, the rate requirement will be set higher.